

THE PUMPING LEMMA - For Regular Languages

Suppose L is a language over the alphabet Σ . If L is accepted by finite automaton $M = (Q, \Sigma, q_0, A, \delta)$, and if P is the number of states of M , then

For every $s \in L$ satisfying $|s| \geq P$, there are three strings x, y and z such that $s = xyz$ and following three conditions are true:

1. $|xy| \leq P$
2. $|y| > 0$ (i.e., $y \neq \epsilon$)
3. For ~~every~~ every $i \geq 0$, the string $xy^iz \in L$

APPLICATION

The most common application of pumping lemma is to show that a language cannot be accepted by an FA by showing that it does not have the property described in the pumping lemma. (We can find ways of applying property described in pumping lemma for a language L that is accepted by an FA)

PROOF BY CONTRADICTION

We assume that for the sake of contradiction, L can be accepted by M , an FA with P states, and we try to select a string in L with length at least P , so that statement 1-3 lead to contradiction.

Why do we Take the Length of the String to Be Greater Than P ?

$$|S| \geq P :$$

Every language accepted by an FA must have one property :

Suppose that $M = (Q, \Sigma, Z_0, A, \delta)$ be an FA accepting language L , and Q has P elements (i.e., P is the number of states of FA).

- If string $S \in L$, having $|S| = P - 1$, then M is in different state after processing every symbol of string S .



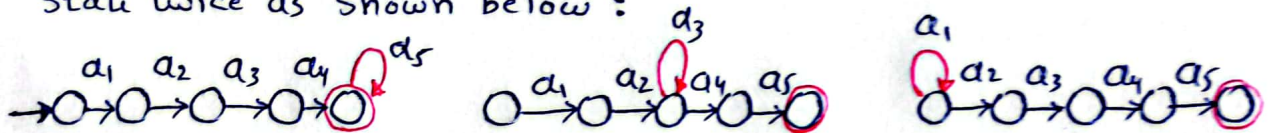
where a_1, a_2, a_3, a_4 are symbols of S

$$S = a_1 a_2 a_3 a_4$$

$$|S| = P - 1 \text{ (here } P = 5)$$

- But if $|S| \geq P$, then

M has read symbols of S , it must have entered some state twice as shown below :



Satisfying $|S| \geq P$, there is some way of breaking S into substrings x, y , and z i.e., M moves from initial state to a final state by following a path that contains a loop.



$$S = xyz$$

NOTE :

Every language accepted by an FA must have a loop in case of infinite language. To ensure existence of loop, we take $|S| \geq P$.

(In case of finite language, there is no need to apply pumping lemma, as all finite languages are regular.)

What is P

We must have a string $S \in L$ with $|S| \geq P$. What is P ?

It is the number of states in finite automata but **we don't know** What is M ? (the whole point is to show that it can't exist).

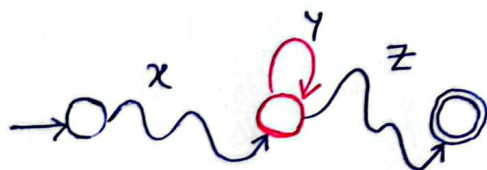
To meet properties of pumping Lemma, choice of S must involve P . we can't say

Let $S = 000111$ (In case of Language $0^n 1^n : n \geq 0$)

Instead, we might say

Let $S = 0^P 1^P$ (as our requirement is that $|S| \geq P$)
or Let $S = 0^{\lceil P/2 \rceil} 1^{\lceil P/2 \rceil}$

$$\underline{|xy| \leq P \text{ and } |y| > 0}$$



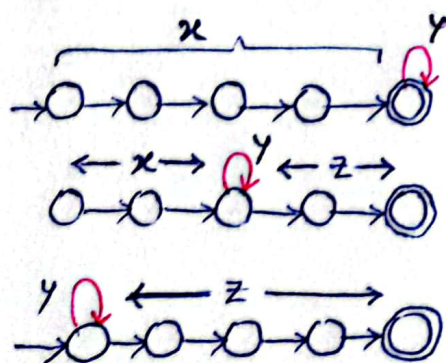
As satisfying $|S| \geq P$, there is some way of breaking S into substring x (before the loop, could be Λ), y (the loop, cannot be Λ) and z (after the loop, could also be Λ).

At least one of the loops (there may be loops) must have completed.
In other words

$$|y| > 0 \quad (\text{i.e., } y \neq \Lambda)$$

Also,

$|xy| \leq P$, as y corresponds to a loop, shown below



(in this case $z = \Lambda$)

(in this case $x = \Lambda$)

But
 $y \neq \Lambda$
in any case

$$\underline{x y^i z \in L \text{ for } i=0,1,\dots,n}$$

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As we know for $|S| \geq p$, at least one of the choices x, y and z , y corresponds to a loop, then there must be more strings that are accepted by M (by executing loop more times) and are therefore in L i.e.,

Strings that cause M to follow the same path but traverse the loop a different number of times.

If $xyz \in L$, then $xy^i z \in L$ for $i=0,1,2,\dots$

$i=0$

If $xyz \in L$, then

$$xy^0 z = xz \in L$$

($y^0 \neq \Lambda$, this shows loop is there but we are not going through it)



$i=1$

$xy^1 z \in L$ (traverse the loop single time)

$i \geq 2$

For each $i \geq 2$, the string $xy^i z \in L$, because M can take the loop i times before proceeding to accept state (still $xyyz, xyyyz$ etc. belong to L)



NOTE

- We must show that there is no value for y that works. Consider all logically possible values for y . then show that each of them fails to satisfy at least one of the 3 statements.
- It is sufficient to find a single value of i such that the resulting string is not in L .

PUMPING LEMMA - EXAMPLES

Question # 1

Using pumping lemma, show that the following language is not regular.

$$A = \{0^n 1^n : n \geq 0\}.$$

Solution:

If A were regular, then there would exist some p such that any string s , where $|s| \geq p$, must satisfy the conditions of the theorem. We show one string s that does not.

Let $s = 0^p 1^p$ $|s| = 2p$, string s is long enough and it is in L .

The string s must satisfy the conditions of the theorem.

So, there must exist x , y , and z , such that $s = x y z$ and

- (i) $|x y| \leq p$
- (ii) $|y| > 0$ (or $y \neq \epsilon$)
- (iii) $\forall i \geq 0, x y^i z \in A$

$$s = 0^p 1^p$$

As $|x y| \leq p$, y must occur within first p characters, i.e.,

$$y = 0^k, \text{ for some } k$$

As $y \neq \epsilon$, p must be greater than 0. Let $i=2$ (pumping one extra copy of y)

$$s = 0^p 1^p$$

$$s = 0^{p+k} 1^p \notin A$$

$$s = 00001111 \in A$$

xyz

$$s = 00 \ 00 \ 1111$$

$x \quad y \quad z$

$$= 00 \ 0000 \ 1111 \notin A$$

$x \quad y \ y \quad z$

Violation of condition 3,

There exists one long string in A that fails to satisfy the condition 3. So the language A is non-regular

CHOICE OF STRING S :

We must have a string $S \in L$ with $|S| \geq p$.

What is p ?

It is the number of states in finite Automata but we don't know what is M - (the whole point is to show that it can't exist).

Choice of S must involve p .

We can't say (in case of Q#1)

$$\text{Let } S = 000111$$

Instead, we might say

$$\text{Let } S = 0^p 1^p \text{ or } \text{Let } S = 0^{\lceil p/2 \rceil} 1^{\lceil p/2 \rceil}$$

here $|S| = p+1$

(as our requirement is that $|S| \geq p$)

Question # 1 (Alternative Method)

Let $s = 0^{\lceil p/2 \rceil} 1^{\lceil p/2 \rceil}$ - $|s| = p+1$, (because of ceiling function), String s is long enough and it is in L
As $|xy| \leq p$, so there are 3 choices of y , only 0's, only 1's, and 01(interleaved).

NOTE 1:

We must show that

There is no value for y that works. To do that, consider all logically possible class of values for y (sometimes there is only one such class, but sometimes several must be considered). Then, we show that each of them fails to satisfy at least one of the 3 conditions.

Case 1: (y as only 0's)

$$xyz = 0^{\lceil p/2 \rceil} 1^{\lceil p/2 \rceil}$$

we take $y=0^k$ for some k

$$xyy = 0^{\lceil p/2 \rceil + k} 1^{\lceil p/2 \rceil} \notin A \quad - \text{ more 0's than 1's}$$

$$xyz = 0^{\lceil p/2 \rceil} 1^{\lceil p/2 \rceil}$$

$$= 000 \ 111 \quad \text{taking } p=5$$

$$= 0 \ 00 \ 111$$

$$\begin{array}{ccc} x & y & z \end{array}$$

$$xyy = 0 \ 0000 \ 111 \notin A$$

Case 2: (y as only 1's)

$$xyz = 0^{\lceil p/2 \rceil} 1^{\lceil p/2 \rceil}$$

we take $y=1^k$ for some k

$$xyy = 0^{\lceil p/2 \rceil} 1^{\lceil p/2 \rceil + k} \notin A \quad - \text{ more 1's than 0's}$$

$$xyz = 0^{\lceil p/2 \rceil} 1^{\lceil p/2 \rceil}$$

$$= 000 \ 111 \quad \text{taking } p=5$$

$$= 000 \ 1 \ 11$$

$$\begin{array}{ccc} x & y & z \end{array}$$

$$xyy = 000 \ 11 \ 11 \notin A$$

Case 3: (y as 01 - interleaved)

$$xyz = 0^{\lceil p/2 \rceil} 1^{\lceil p/2 \rceil}$$

we take $y=01$

$$xyy = 0^{\lceil p/2 \rceil - 1} 0101 \ 1^{\lceil p/2 \rceil - 1} \notin A \quad - \text{ 01 interleaved}$$

$$xyz = 0^{\lceil p/2 \rceil} 1^{\lceil p/2 \rceil}$$

$$= 000 \ 111 \quad \text{taking } p=5$$

$$= 00 \ 01 \ 11$$

$$\begin{array}{ccc} x & y & z \end{array}$$

$$xyy = 00 \ 0101 \ 11 \notin A$$

NOTE 2:

To make maximum use of pumping theorem's requirement that y fall in the first p characters, it is a good idea to choose a string s longer than p characters so that there is a uniform first region of length at least p . It may be possible to consider just a single case for where y can fall. Choose

$$s = 0^p 1^p \text{ - as shown in previous page}$$

NOTE 3:

It suffices to find a single long string s that is in L but not pumpable.

NOTE 4:

Generally, we do that by assuming that y does satisfy the first 2 conditions, then we consider the 3rd condition to show that it is not possible to satisfy that requirement.

NOTE 5:

It is sufficient to find a single value of i such that the resulting string is not in L (typically, this can be done by setting i to 0 (pumping out once) or to 2 (pumping once). Sometimes other value of i must be considered.

Question # 2

Using pumping lemma, show that the following language is not regular.

$$F = \{ ww : w \in \{0, 1\}^* \}.$$

Solution:

$$\text{Let } s = 0^p 0^p$$

As $|xy| \leq p$, y must occur within first p characters, i.e.,

$$y = 0$$

$$s = 000\ 000 \in F \quad (\text{taking } p=3)$$

xyz

$$s = 00\ 0\ 000$$

x y z

$$= 00\ 0\ 0\ 000 \notin F \quad (ww \text{ contains even \# of 0's})$$

x y y z

But wait

If we take $y = 00$

$$s = 000\ 000 \in F \quad (\text{taking } p=3)$$

xyz

$$s = 0\ 00\ 000$$

x y z

$$= 0\ 00\ 00\ 000 \in F \quad (ww \text{ contains even \# of 0's})$$

x y y z

Here string is pumpable.

Hence, no violation of condition is observed.

We must show that there is no value for y that works. Using pumping lemma required some skill. So, selecting the string s here required more care.

It suffices to find a single long string s that is in F but not pumpable.

Taking another string from language F , such as

$$\text{Let } s = 0^p 1 0^p 1 \quad |s| = 2p+2, \text{ string } s \text{ is long enough and it is in } F.$$

So, there must exist x , y , and z , such that $s = xyz$ and must satisfy the following conditions.

- (i) $|xy| \leq p$
- (ii) $|y| > 0$ (or $y \neq \epsilon$)
- (iii) $\forall i \geq 0, xy^i z \in F$

$$s = 0^p 1 0^p 1$$

As $|xy| \leq p$, y must occur within first p characters, i.e.,

$$y = 0^k, \text{ for some } k$$

As $y \neq \epsilon$, p must be greater than 0. Let $i=2$ (pumping one extra copy of y)

$$s = 0^p 1 0^p 1$$

$$s = 0^{p+k} 1 0^p 1 \notin F$$

$$s = 0001 \ 0001 \in F$$

xyz

$$s = 0 \ 00 \ 10001$$

$x \quad y \quad z$

$$= 0 \ 0000 \ 10001 \notin F$$

$x \quad y \ y \quad z$

Violation of condition 3,

There exists one long string in F that fails to satisfy the condition 3. So the language F is non-regular.

Question # 3

Using pumping lemma, show that the following language is not regular.

$D = \{ 1^{n^2} : n \geq 0 \}$ - D contains all strings of 1's whose length is perfect square.

Solution:

$D = \{ 1^{n^2} : n \geq 0 \}$ - D contains all strings of 1's whose length is perfect square.

$D = \{ \epsilon, 1, 1111, 1111111111, 1111111111111111, \dots \}$

i.e., # of 1's are 0, 1, 4, 9, 16, 25, 36, 49,

If D were regular, then there would exist some p such that any string s , where $|s| \geq p$, must satisfy the conditions of the theorem. We show one string s that does not.

$$\text{Let } s = 1^{p^2} \quad |s| = p^2$$

So, there must exist x , y , and z , such that $s = xyz$ and must satisfy the following conditions.

- (i) $|xy| \leq p$
- (ii) $|y| > 0$ (or $y \neq \epsilon$)
- (iii) $\forall i \geq 0, xy^i z \in D$

$$s = 1^{p^2}$$

In Set D , # of 1's are 0, 1, 4, 9, 16, 25, 36, 49, 64,

Note the growing gap between successive members (# Of 1's) of this sequence. Large members of this sequence cannot be near each other.

The two strings xyz of D and successive string $xyyz$ differs in length by the length of y . As the length of xyz of D is a perfect square, then the length of its successive string will also be perfect square.

We have

$$|xyz| = p^2 \quad \text{As, } s = xyz = 1^{p^2}$$

The length of the successive string of xyz (i.e., length of $xyyz$) should be perfect square.

But

$$\text{As } |xyz| = p^2$$

We know from the pumping theorem,

$$|xy| \leq p$$

$$\text{Also } |y| \leq p$$

Then

$$|xyyz| \leq p^2 + p \quad (\text{pumping once, operator = replaced by } \leq, \text{ because } |y| \leq p)$$

$$|xyyz| \leq p^2 + p$$

$$\text{As we know } p^2 + p < p^2 + 2p + 1$$

$$\text{or } p^2 + p < (p+1)^2$$

$$|xyyz| \leq p^2 + p \quad \text{---- (A)}$$

$$\text{and } p^2 + p < (p+1)^2 \quad \text{---- (B)}$$

From (A) and (B),

$$|xyyz| < (p+1)^2 \quad \text{-----(1)}$$

$$\text{As } |xyz| = p^2, \text{ so}$$

$$|xyyz| > p^2 \quad (\text{Because } y \neq \epsilon) \quad \text{-----(2)}$$

From (1) and (2),

$$p^2 < |xyyz| < (p+1)^2$$

As length of xyz is perfect square (i.e., p^2) but the length of its successive string $xyyz$ is not perfect square (lies strictly between consecutive perfect squares (p^2 and $(p+1)^2$)). This length cannot be perfect square. So, we arrive at the contradiction,

$$|xyyz| \notin D$$

There exists a string in D that fails to satisfy the condition 3. So the language D is non-regular.

Question # 4

Using pumping lemma, show that the following language is not regular.

$$E = \{ w \in \{ a, b \}^* : \#_a(w) = \#_b(w) \}.$$

Solution:

Sometimes the easiest way to prove that a language is not regular is to use the closure properties for regular languages, either alone or in conjunction with pumping theorem.

Let E were regular, then

$L = E \cap a^*b^*$ would also be regular - (According to closure properties of regular).

But $L = E \cap a^*b^* = a^n b^n, n \geq 0$

We know $a^n b^n, n \geq 0$ is non-regular,

So E will be non-regular.

Question # 5

Using pumping lemma, show that the following language is not regular.

$$A = \{ 0^i 1^j : i > j \}$$

Solution:

Assume that A is regular. Let p be the pumping length for A given by the pumping lemma.

$$\text{Let } s = 0^{p+1} 1^p.$$

Then s can be split into xyz , satisfying the conditions of the pumping lemma.

As $|xy| \leq p$, y consists only of 0's.

Let's examine the string $xyyz$ to see whether it can be in A . Adding an extra copy of y increases the number of 0's. But, A contains all strings in 0^*1^* that have more 0's than 1's, so increasing the number of 0's will still give a string in A . No contradiction occurs.

We need to try something else.

The pumping lemma states that $\forall i \geq 0, xy^i z \in A$, even when $i = 0$,

so let's consider the string $xy^0 z = xz$. Removing string y decreases the number of 0's in s .

Recall that s has just one more 0 than 1. Therefore xz cannot have more 0's than 1's, so it cannot be a member of A . Thus we obtain a contradiction.

ACTIVITY # 4 (SOLUTION)

Question # 1

Using pumping lemma, show that the following language is not regular.

Bal = {w ∈ {), (}* : the parenthesis are balanced}.

e.g. $()$, $(())$, $((()))$, $(((()))$, $((((()))$,

Solution:

If **Bal** were regular, then there would exist some p such that any string s , where $|s| \geq p$, must satisfy the conditions of the theorem. We show one string s that does not.

Bal contains complex strings like $((()))((()))$. But it is almost easier to use pumping lemma if we pick as simple a single as possible. So,

Let $s = (p)^p$ $|s| = 2p$, string s is long enough and it is in **Bal**.

So, there must exist x , y , and z , such that $s = x y z$ and must satisfy the following conditions.

- (i) $|x y| \leq p$
- (ii) $|y| > 0$ (or $y \neq \epsilon$)
- (iii) $\forall i \geq 0, xy^iz \in \text{Bal}$

$$S = \binom{p}{p} p$$

As $|x y| \leq p$, y must occur within first p characters, i.e.,

$$y = \binom{k}{i}, \text{ for some } k$$

As $y \notin \epsilon$, p must be greater than 0. Let $i=2$ (pumping one extra copy of y)

$$S = (p)^p$$

$$S = (p+k)^p \notin \text{Bal}$$

$$\begin{aligned} s &= (((())) \in \text{Bal} \\ &\quad \text{xyz} \\ s &= (((())) \\ &\quad \text{x} \quad \text{y} \quad \text{z} \\ &= ((((())) \notin \text{Bal} \\ &\quad \text{x} \quad \text{y} \quad \text{y} \quad \text{z} \end{aligned}$$

Violation of condition 3,

There exists one long string in **Bal** that fails to satisfy the condition 3. So the language **Bal** is non-regular.

Question # 2

Using pumping lemma, show that the following language is not regular.

$$\text{PalEven} = \{ ww^R : w \in \{a, b\}^* \}.$$

Solution:

If **PalEven** were regular, then there would exist some p such that any string s , where $|s| \geq p$, must satisfy the conditions of the theorem. We show one string s that does not.

Let $s = a^p b^p b^p a^p$ $|s| = 4p$, string s is long enough and it is in **PalEven**.

So, there must exist x , y , and z , such that $s = xyz$ and must satisfy the following conditions.

- (i) $|xy| \leq p$
- (ii) $|y| > 0$ (or $y \neq \epsilon$)
- (iii) $\forall i \geq 0, xy^i z \in \text{PalEven}$

$$s = a^p b^p b^p a^p$$

As $|xy| \leq p$, y must occur within first p characters, i.e.,

$$y = a^k, \text{ for some } k$$

As $y \neq \epsilon$, p must be greater than 0. Let $i=2$ (pumping one extra copy of y)

$$s = a^p b^p b^p a^p$$

$$s = a^{p+k} b^p b^p a^p \notin \text{PalEven}$$

$$s = aaabbb bbbaaa \in \text{PalEven}$$

$$xyz$$

$$s = a \quad aa \quad bbb \quad bbbaaa$$

$$x \quad y \quad z$$

$$= a \quad aa \quad aa \quad bbb \quad bbbaaa \notin \text{PalEven}$$

$$x \quad y \quad y \quad z$$

Violation of condition 3,

There exists one long string in **PalEven** that fails to satisfy the condition 3. So the language **PalEven** is non-regular.

Question # 3

Using pumping lemma, show that the following language is not regular.

$$L = \{a^i b^j : i, j \geq 0 \text{ and } i \neq j\}.$$

Solution:

The closure theorem provides an easier way to prove that L language is not regular. -

Let L were regular, then

We observe that if L were regular, then $\neg L$ would also be regular, since the regular languages are closed under complement.

As
$$L = \{a^i b^j : i, j \geq 0 \text{ and } i \neq j\}.$$

$$\neg L = \{a^n b^n : n > 0\} \cup \{\text{strings of a's and b's that do not have all a's in front of all b's}\}$$

If $\neg L$ is regular, then $\neg L \cap a^*b^*$ must also be regular.

But

$$\neg L \cap a^*b^* = \{a^n b^n : n > 0\}, \text{ which we have already shown is not regular.}$$

So, neither is $\neg L$ or L is regular.

Question # 4

Using pumping lemma, show that the following language is not regular.

$$L = \{a^i b^j c^k : i, j, k \geq 0 \text{ and (if } i = 1 \text{ then } j = k)\}.$$

Solution:

Let $L = \{a^i b^j c^k : i, j, k \geq 0 \text{ and (if } i = 1 \text{ then } j = k)\}.$

Observations:

It is easier to see this if we rewrite the final condition as $(i \neq 1) \text{ or } (j = k)$.

• If $i = 0$ then:

If $j \neq 0$, let y be b ; otherwise, let y be c .

Pump in or out. Then i will still be 0 and thus not equal to 1, so the resulting string is in L .

• If $i = 1$ then:

Let y be a . Pump in or out. Then i will no longer equal 1, so the resulting string is in L .

• If $i = 2$ then:

Let y be aa . Pump in or out. Then i cannot equal 1, so the resulting string is in L .

• If $i > 2$ then:

Let y be a . Pump out once or in any number of times. Then i cannot equal 1, so the resulting string is in L .

But L is not regular.

One way to prove this is to use the fact that the regular languages are closed under intersection.

$$L = \{a^i b^j c^k : i, j, k \geq 0 \text{ and (if } i = 1 \text{ then } j = k)\}.$$

if L were regular,

then

$$\neg L = L \cap ab^*c^* = \{a b^j c^k : j, k \geq 0 \text{ and } j = k\} \text{ would also be regular.}$$

$$\text{Let } w = a b^p c^p.$$

Then y must occur in the first p characters of w .

If y includes the initial a ,

Pump in once, the resulting string is not in $\neg L$ because it contains more than one a .

If y does not include the initial a , then it must be b^k , where $0 < k < p$.

Pump in once, the resulting string is not in $\neg L$ because it contains more b 's than c 's.

So, $\neg L$ is non-regular and according to closure property, L is also non-regular.